



**Yashwantrao Chavan Institute of Science,
Satara (Autonomous)**

Department of Computer Science (Entire)

B.Sc.COMPUTER

SCIENCE (ENTIRE) – III (2025-2026)

Project On

Student Marketplace

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Examiner

Head of Department

Acknowledgements

A good project is result of lot of efforts of many people who are involved directly or indirectly. We take opportunity to gracefully thank to all those people without whom this project would not have succeeded. Each of them has their own special way contributed towards the success of this project.

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We are also honestly thankful for our all friends who gives us their valuable suggestions about our project and help us to complete this project.



Roll No.:

UID No.:

“Education through self-help is our motto”-Karmaveer.

Rayat Shikshan Sanstha's

**YASHAVANTRAO CHAVAN INSTITUTE OF SCIENCE, SATARA
(AUTONOMOUS)**

CERTIFICATE

DEPARTMENT OF COMPUTER SCIENCE(ENTIRE)

B.Sc. Computer Science (Entire)-Part-III

Date :- - -2025

This is to certify that partial fulfillment of curriculum of T.Y.B.C.S. Student

Mr.Aditya Shrimant Gaikwad & Pavan Sunil Gaikwad has successfully completed the project Working the Computer Science entitled **“Student Marketplace”** prescribed by The **KBP University** and this project report represents Her bonafide work in year **2025-2026**.

PROJECT GUIDE

EXAMINER

HEAD ODEPARTMENT

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Introduction

The Student Marketplace is a web-based ecommerce platform developed to facilitate buying and selling products among college students. Many students need academic materials and personal items temporarily, and purchasing brand-new products can be costly. This platform helps students access affordable resources by enabling them to sell and buy used items within their campus community.

Objective

To provide an affordable platform for students to buy and sell goods.

To reduce waste by promoting reuse of student items.

To create a secure, user-friendly ecommerce system for students.

To enable easy communication between buyers and sellers.

To support fast searching and categorized browsing of products.

Scope

User Registration & Login for Students and Admin.

Students can upload items with details and images.

Buyers can view product listings and send purchase requests.

Admin can manage users, product approvals, and reports.

Product search by category, price, and location.

Order management and notifications to users.

Project Plan

Milestones

Phases involved:

1. Requirement Analysis
2. System Design
3. Database Design
4. Frontend Development (HTML, CSS, JS)
5. Backend Development (Python Flask)
6. Integration with MySQL Database
7. Testing and Debugging
8. Deployment & Maintenance

Existing System

Traditionally, students rely on:

Social media buy/sell groups

Word of mouth

Offline notice boards

These systems are unorganized, insecure, and not centrally managed.

Background of Organization

The platform is developed for college/university students to exchange goods among themselves. It aims to improve resource access on campus and minimize unnecessary expenses. The system is designed keeping the student environment and needs in mind.

Current Problems

Lack of trustworthy and centralized marketplace for students.

No proper product categorization.

High product cost when bought new.

Risk of fraud in offline deals.

Difficulty in finding buyers/sellers quickly.

Areas for Improvement

Improve product verification process.

Introduce delivery system inside campus.

Implement product rating and review system.

Payment gateway integration for secure transactions.

Proposed System

The proposed Student Marketplace:

Provides secure login and user authentication.

Allows easy posting of product details with images.

Maintains a well-organized category-based product listing.

Enables users to send and manage purchase requests.

Admin panel supports moderation and product approval.

Improves communication between users.

Input/Output Requirements

Input

User login details

Product details (name, price, description, category, image)

Order requests

✓ Output

Successful product listing

Display of available products

Notifications and order confirmations

Admin approvals and reports

Hardware and Software Requirements

Category	Requirements
Hardware	PC/Laptop, 4GB+ RAM, 500GB Storage
Operating System	Windows / Linux
Frontend	HTML, CSS, JavaScript
Backend	Python Flask
Database	MySQL
Browser	Chrome, Firefox, Edge

Database Requirement

Database Name: student_marketplace

Tables:

users

products

categories

orders

admin

Relations established using primary & foreign keys.

System Design

Database Dictionary (Example)

Table	Field	Type	Description
users	user_id	INT (PK)	Unique Student ID
	name	VARCHAR	Student Name
	email	VARCHAR	Login Email
	password	VARCHAR	Encrypted Password
products	product_id	INT (PK)	Unique Product ID
	user_id	INT (FK)	Seller ID
	product_name	VARCHAR	Item Name
	price	INT	Product Price
	description	TEXT	Details
	category	VARCHAR	Product Category
orders	order_id	INT (PK)	Order Request ID
	product_id	INT (FK)	Ordered Product

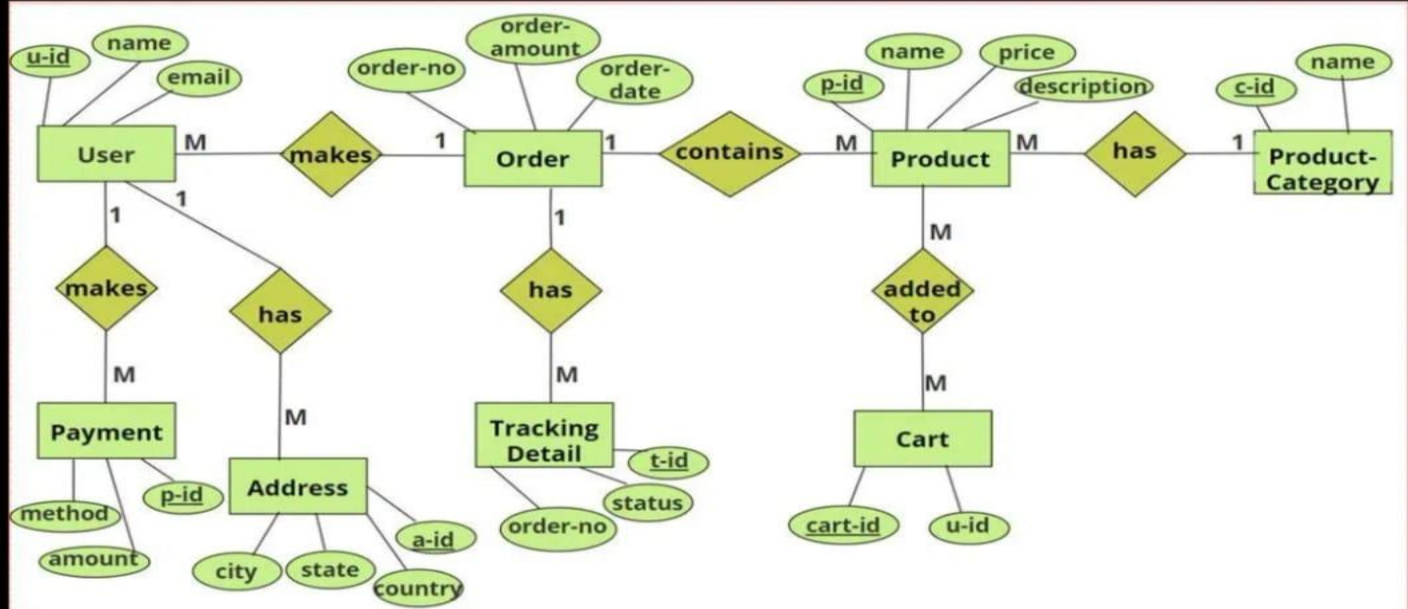
Database Dictionary

Products Table — Example Record

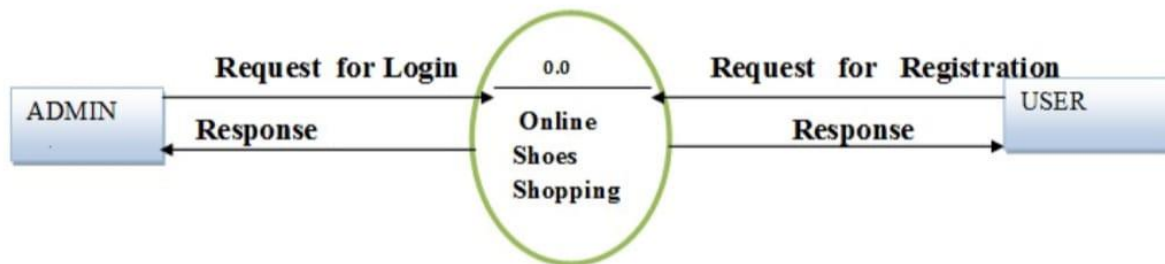
Field Example Value

product_id 301
user_id 101
product_name Data Structures Book
category Books
price 250.00
description Good condition, 2nd year syllabus
image /uploads/book1.jpg
status Active
created_at 2025-02-02 12:10:00

ER Diagram

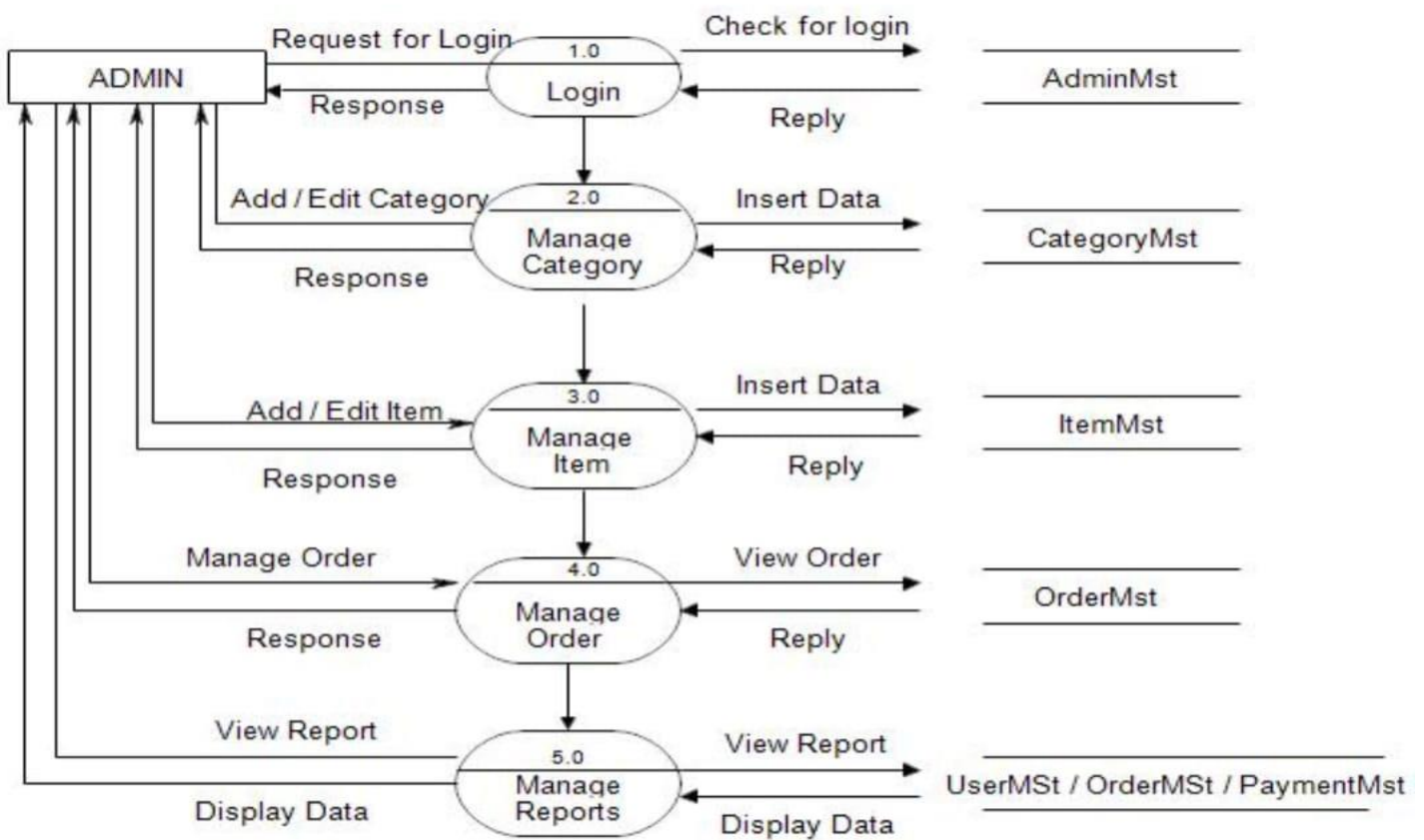


Level 0 DFD



Level 1 DFD

Admin Side DFD - 1st Level



Security Controls

Password encryption using hashing (bcrypt).

Session-based authentication in Flask.

Role-based access: Admin vs User.

Access control to prevent unauthorized product editing.

Data validation & SQL injection prevention.

Future Scope of the Project

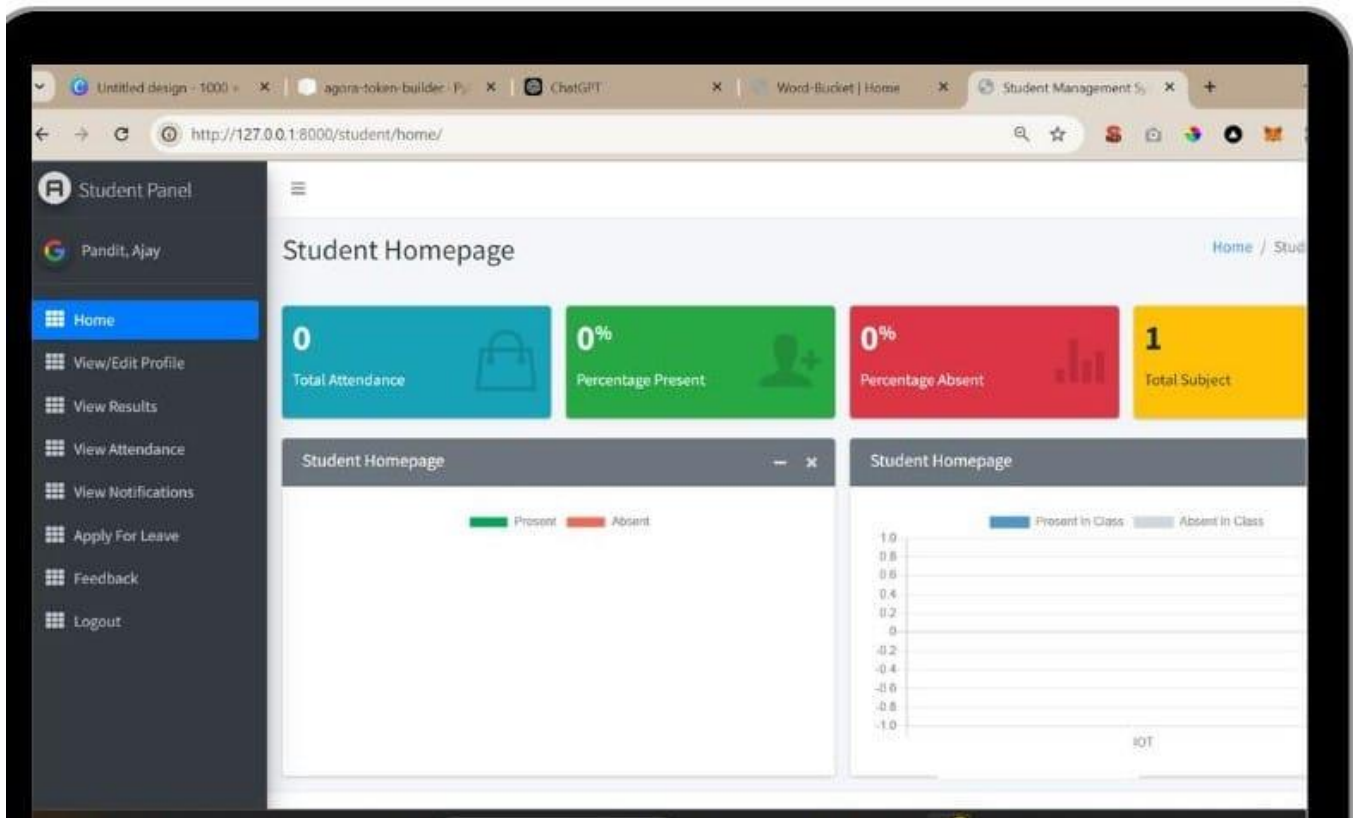
Online payment gateway integration.

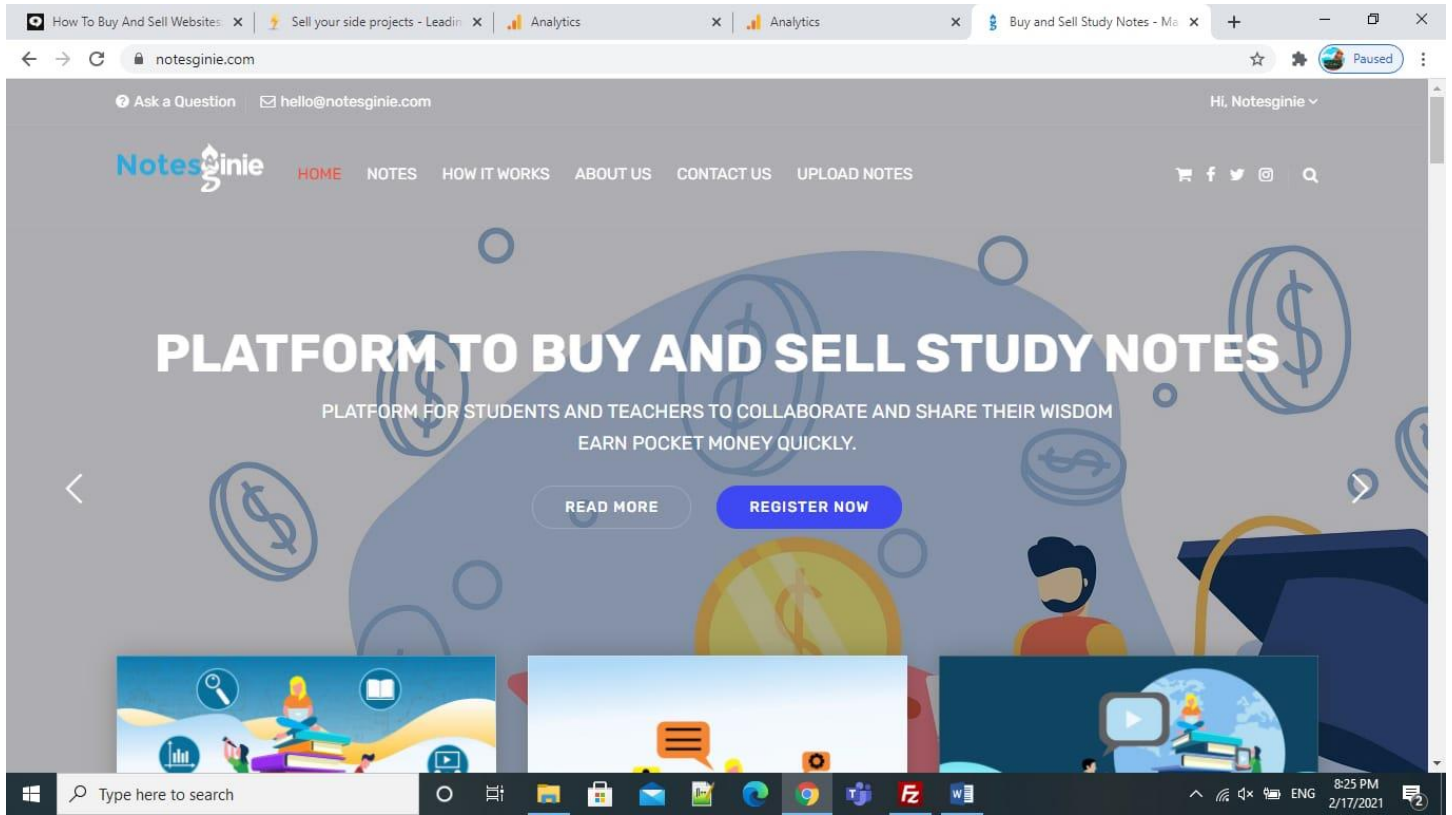
Chat system between buyers and sellers.

Mobile Application development.

AI-based product recommendations.

Delivery & pickup tracking inside campus.





Conclusion

The Student Marketplace project successfully provides a safe, low-cost, and student-friendly ecommerce platform that improves accessibility of resources within the campus. It overcomes the drawbacks of traditional systems and enables smooth buying and selling of pre-owned products.

References

Flask Official Documentation

MySQL Developer Guide

W3Schools HTML, CSS, JavaScript

Python.org Tutorials